Prateek Goswami

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Summary

Computer Science & AI undergraduate with hands-on experience building end-to-end ML pipelines, AI-powered tools, and multimodal systems. Proficient in Python, agentic AI workflows, and deploying models across NLP, computer vision, and large-scale data tasks. Excited to contribute to AI agent development and internal tooling at a fast-growing health-tech startup.

Education

JK Lakshmipat University Jaipur, Rajasthan
B.Tech in Computer Science & Artificial Intelligence — CGPA: 9.09/10 *2023 – Present*
IIT Gandhinagar (Exchange Semester 3) Gandhinagar, Gujarat
Coursework: Computer Networks & Security, Foundations of AI
IIT Jammu (Exchange Semester 5) Jammu, J&K
Coursework: Deep Learning, Computer Vision, Artificial Intelligence, DBMS

Experience

AI / ML Engineer Intern Remote
Patidar Agriculture — AgriTech (Agricultural Machinery) *Jun 2024 – Jul 2024*

- Built and deployed an AI-powered chatbot for the company website to handle product queries, machinery recommendations, and customer support — integrating an LLM backend with a conversational interface.
- Developed computer vision models for disease detection using image processing and predictive models for data-driven decision support using Python, OpenCV, and Scikit-learn.
- Worked across the full ML lifecycle — data collection, preprocessing, model training, evaluation, and deployment — delivering end-to-end solutions for a production website used by real customers.

Skills

Languages: Python, JavaScript, C, C++, SQL

AI / Agentic: LLM APIs (Claude, OpenAI), AI Agent frameworks, Prompt Engineering, Tool Use, RAG pipelines

ML / DL: Scikit-learn, TensorFlow, PyTorch, Keras, HuggingFace Transformers, PySpark ML, MLflow

CV / NLP: OpenCV, Mediapipe, YOLOv5, Object Detection, Image Captioning, VQA, Sentiment Analysis

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Spark SQL, EDA, Model Evaluation

Tools: Databricks, Docker, GitHub, Jupyter Notebook, Google Colab, Figma

Projects

AI Agent & Internal Tool Development | *Python, LLM APIs, Claude, Tool Use*

- Designed and deployed AI agents leveraging Claude and OpenAI APIs with tool-use/function-calling capabilities to automate multi-step workflows and productivity tasks.
- Built internal tooling integrating LLMs with external APIs and data sources; implemented prompt engineering strategies (chain-of-thought, few-shot) to maximise agent reliability.
- Developed agent evaluation loops — monitoring output quality, detecting hallucinations, and iterating on system prompts to optimise task completion rates.

Real-Time Anti-Cheating Proctoring System | *Python, YOLOv5, Mediapipe, TensorFlow, OpenCV*

- Built a real-time multi-model inference system detecting gaze deviation, head pose, hand gestures, and unauthorised devices; logged behaviour events with timestamped alerts for automated monitoring.
- Integrated and orchestrated multiple AI models in a single pipeline — demonstrating applied agentic system design with real-world deployment constraints.

Customer Churn Prediction Pipeline | *PySpark, Databricks, MLflow, Scikit-learn*

- Built an end-to-end binary classification pipeline on Databricks using PySpark for large-scale data processing and Spark SQL for EDA across 7,000+ customer records.
- Engineered features via StringIndexer and VectorAssembler; trained Logistic Regression and Random Forest classifiers; tracked experiments with MLflow. Achieved AUC-ROC of 0.83 and 78.4% accuracy.

Visual Question Answering (VQA) System | *PyTorch, ResNet/ViT, BERT, Transformers*

- Built a multimodal AI model combining image feature extraction (ResNet/ViT) with question encoding (BERT) to generate natural language answers; evaluated on the VQA v2 benchmark across open-ended and binary question types.

Sentiment Analysis Pipeline | *Python, HuggingFace Transformers, Scikit-learn, NLP*

- Developed an end-to-end NLP pipeline for multi-class sentiment classification; compared TF-IDF + classical ML baselines with fine-tuned transformer models; reported F1-score and confusion matrix evaluation.

Image Captioning Model | *TensorFlow, CNN-LSTM, Attention, Flickr30k*

- Developed a CNN-LSTM captioning pipeline with Bahdanau attention trained on Flickr30k; improved BLEU-4 score through attention integration and systematic hyperparameter tuning.

Lip Sync Generation / Audio-Visual Synchronisation | *Python, Wav2Lip, FFmpeg*

- Fine-tuned a Wav2Lip-based architecture to synchronise lip movements in video with arbitrary audio; built a preprocessing pipeline covering face detection, mouth-ROI extraction, and audio-video alignment.

Leadership & Achievements

Organising Head — College Hackathon JK Lakshmipat University
Led a cross-functional team of 200+ volunteers across logistics, scheduling, and event execution *2026*

- Coordinated sponsor outreach, participant onboarding, and real-time problem resolution across all hackathon tracks.

Teaching Assistant — C Programming JK Lakshmipat University
Selected based on academic performance and programming expertise to support 60+ students *2025*

- Conducted doubt sessions, graded assignments, debugging, and core C topics.

Dean's Honour List JK Lakshmipat University
Recognised for outstanding academic performance across consecutive semesters